

Homework 5
Math. 481a, Spring 2026
Prepared in L^AT_EX

Problem 1.

Claim. If $f''(x_0)$ exists and f is defined in some open interval containing x_0 , then

$$\lim_{h \rightarrow 0} \frac{1}{h^2} [f(x_0 - h) - 2f(x_0) + f(x_0 + h)] = f''(x_0).$$

Proof. Since $f''(x_0)$ exists, f' exists on some open interval $(x_0 - \delta, x_0 + \delta)$, and by definition of the second derivative,

$$\lim_{h \rightarrow 0} \frac{f'(x_0 + h) - f'(x_0)}{h} = f''(x_0), \quad \lim_{h \rightarrow 0} \frac{f'(x_0 - h) - f'(x_0)}{-h} = f''(x_0).$$

Define $\varphi(h) = f(x_0 + h) - 2f(x_0) + f(x_0 - h)$. Then $\varphi(0) = 0$, and for $|h| < \delta$, φ is differentiable in h with

$$\varphi'(h) = f'(x_0 + h) - f'(x_0 - h).$$

The numerator $\varphi(h)$ and denominator h^2 both vanish at $h = 0$, and $\frac{d}{dh}(h^2) = 2h \neq 0$ for $h \neq 0$. Apply L'Hôpital's rule once:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\varphi(h)}{h^2} &= \lim_{h \rightarrow 0} \frac{f'(x_0 + h) - f'(x_0 - h)}{2h} = \frac{1}{2} \lim_{h \rightarrow 0} \left[\frac{f'(x_0 + h) - f'(x_0)}{h} + \frac{f'(x_0 - h) - f'(x_0)}{-h} \right] \\ &= \frac{1}{2} [f''(x_0) + f''(x_0)] = f''(x_0), \end{aligned}$$

where the split into two difference quotients is valid because each individual limit exists ($= f''(x_0)$).

$$\lim_{h \rightarrow 0} \frac{1}{h^2} [f(x_0 - h) - 2f(x_0) + f(x_0 + h)] = f''(x_0).$$

□

Problem 2.

$$I = \int_1^{1.5} x^2 \ln x \, dx, \quad f(x) = x^2 \ln x.$$

Derivatives:

$$f''(x) = 2 \ln x + 3, \quad f^{(4)}(x) = -\frac{2}{x^2}.$$

Exact value. Integrating by parts ($u = \ln x$, $dv = x^2 \, dx$):

$$\begin{aligned} \int x^2 \ln x \, dx &= \frac{x^3}{3} \ln x - \frac{x^3}{9} + C \\ I &= \left[\frac{(1.5)^3}{3} \ln 1.5 - \frac{(1.5)^3}{9} \right] - \left[0 - \frac{1}{9} \right] = \frac{9}{8} \ln \frac{3}{2} - \frac{19}{72} \approx 0.1922593577. \end{aligned}$$

Trapezoidal rule. $h = b - a = 0.5$, $x_0 = 1$, $x_1 = 1.5$:

$$T = \frac{h}{2}[f(x_0) + f(x_1)] = \frac{0.5}{2}\left[0 + \frac{9}{4}\ln\frac{3}{2}\right] = \frac{9}{16}\ln\frac{3}{2} \approx 0.2280741.$$

Actual error: $|I - T| \approx 3.5815 \times 10^{-2}$.

Error bound (with $h = b - a$): $|E_T| \leq \frac{h^3}{12} \max_{[1,1.5]} |f''|$. Since f'' is increasing, $\max |f''| = f''(1.5) = 2\ln 1.5 + 3 \approx 3.8109$.

$$|E_T| \leq \frac{(0.5)^3}{12}(2\ln 1.5 + 3) \approx 3.9697 \times 10^{-2}.$$

Simpson's rule. $h = 0.25$, $x_0 = 1$, $x_1 = 1.25$, $x_2 = 1.5$:

$$S = \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)] = \frac{1}{12}[0 + 4(1.5625)\ln 1.25 + 2.25\ln 1.5] \approx 0.1922453.$$

Actual error: $|I - S| \approx 1.4050 \times 10^{-5}$.

Error bound: $|E_S| \leq \frac{h^5}{90} \max_{[1,1.5]} |f^{(4)}|$. Since $|f^{(4)}(x)| = 2/x^2$ is decreasing, $\max |f^{(4)}| = |f^{(4)}(1)| = 2$.

$$|E_S| \leq \frac{(0.25)^5}{90}(2) \approx 2.1701 \times 10^{-5}.$$

Comparison.

Rule	Approximation	Actual error	Error bound
Trapezoidal	0.22807412	3.5815×10^{-2}	3.9697×10^{-2}
Simpson's	0.19224531	1.4050×10^{-5}	2.1701×10^{-5}

In both cases, $|\text{actual error}| \leq |\text{error bound}|$, as required.

Problem 3.

Let $x_1 = x_0 + h$, $x_2 = x_0 + 2h$. We seek a_0, a_1, a_2, k such that

$$\int_{x_0}^{x_2} f(x) dx = a_0 f(x_0) + a_1 f(x_1) + a_2 f(x_2) + k f^{(4)}(\xi).$$

The coefficients a_i are independent of x_0 , so each test polynomial $f(x) = x^n$ must give an identity in x_0 . Equate coefficients of powers of x_0 .

$n = 1$: $f(x) = x$, $f^{(4)} \equiv 0$.

$$\int_{x_0}^{x_0+2h} x dx = 2x_0 h + 2h^2, \quad \sum a_i(x_0 + ih) = (a_0 + a_1 + a_2)x_0 + (a_1 + 2a_2)h.$$

$$a_0 + a_1 + a_2 = 2h, \quad a_1 + 2a_2 = 2h. \quad (\text{I, II})$$

$n = 2$: $f(x) = x^2$, $f^{(4)} \equiv 0$.

$$\int_{x_0}^{x_0+2h} x^2 dx = 2x_0^2 h + 4x_0 h^2 + \frac{8}{3}h^3,$$

$$\sum a_i(x_0 + ih)^2 = (a_0 + a_1 + a_2)x_0^2 + (2a_1 + 4a_2)x_0h + (a_1 + 4a_2)h^2.$$

The coefficients of x_0^2 and x_0 reproduce (I), (II); the constant term gives:

$$a_1 + 4a_2 = \frac{8}{3}h. \quad (\text{III})$$

$n = 3$: expanding $\sum a_i(x_0 + ih)^3$ produces only one new equation (the constant-in- x_0 coefficient): $a_1 + 8a_2 = 4h$. With $a_1 = 4h/3$ and $a_2 = h/3$ obtained below, $4h/3 + 8h/3 = 4h$ holds automatically. So $n = 3$ contributes no new constraint, and Simpson's rule has degree of precision ≥ 3 .

Solving (I)–(III): (III) – (II) $\Rightarrow 2a_2 = 2h/3 \Rightarrow a_2 = h/3$; then (II) $\Rightarrow a_1 = 2h - 2(h/3) = 4h/3$; and (I) $\Rightarrow a_0 = 2h - 4h/3 - h/3 = h/3$.

$$\boxed{a_0 = \frac{h}{3}, \quad a_1 = \frac{4h}{3}, \quad a_2 = \frac{h}{3}.}$$

Finding k . Apply to $f(x) = x^4$, for which $f^{(4)}(x) = 24$ (constant, so $f^{(4)}(\xi) = 24$). Take $x_0 = 0$ (translation invariance):

$$\begin{aligned} \text{LHS} &= \int_0^{2h} x^4 dx = \frac{32h^5}{5}, \\ \text{RHS} &= \frac{h}{3} \cdot 0 + \frac{4h}{3} \cdot h^4 + \frac{h}{3} \cdot 16h^4 + 24k = \frac{4h^5}{3} + \frac{16h^5}{3} + 24k = \frac{20h^5}{3} + 24k. \\ \frac{32h^5}{5} &= \frac{20h^5}{3} + 24k \implies 24k = \frac{96h^5 - 100h^5}{15} = -\frac{4h^5}{15} \implies \boxed{k = -\frac{h^5}{90}}. \end{aligned}$$

Simpson's rule with error term:

$$\boxed{\int_{x_0}^{x_2} f(x) dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90} f^{(4)}(\xi).}$$

Problem 4.

Definition 4.1 (textbook, p. 195): The degree of precision of a quadrature formula is the largest positive integer n such that the formula is exact for x^k for each $k = 0, 1, \dots, n$.

Let $E(f) = \int_a^b f(x) dx - \sum_{i=0}^m a_i f(x_i)$ denote the error functional. Both the integral and the finite sum are linear in f , so

$$E[\alpha f + \beta g] = \alpha E[f] + \beta E[g], \quad \forall \alpha, \beta \in \mathbb{R},$$

i.e., E is linear. By definition, “the formula is exact for f ” $\iff E(f) = 0$.

Claim: The formula has degree of precision n if and only if $E(x^k) = 0$ for all $k = 0, 1, \dots, n$ and $E(x^{n+1}) \neq 0$.

Proof.

($\implies$) Suppose the formula has degree of precision n . By Definition 4.1, the formula is exact for x^k for each $k = 0, 1, \dots, n$; that is,

$$E(x^k) = 0, \quad k = 0, 1, \dots, n.$$

Suppose for contradiction that $E(x^{n+1}) = 0$. Then for every polynomial $P(x) = \sum_{k=0}^{n+1} c_k x^k$ of degree $\leq n+1$, linearity gives

$$E(P) = \sum_{k=0}^{n+1} c_k E(x^k) = 0,$$

so the formula is exact for every polynomial of degree $\leq n+1$ — in particular for x^{n+1} . This contradicts the maximality of n . Hence $E(x^{n+1}) \neq 0$.

($\Leftarrow$) Suppose $E(x^k) = 0$ for $k = 0, 1, \dots, n$ and $E(x^{n+1}) \neq 0$.

For any polynomial P of degree $\leq n$, write $P(x) = \sum_{k=0}^n c_k x^k$. By linearity,

$$E(P) = \sum_{k=0}^n c_k E(x^k) = 0,$$

so the formula is exact for every polynomial of degree $\leq n$.

The polynomial x^{n+1} has degree $n+1$, and $E(x^{n+1}) \neq 0$ shows the formula is not exact for it. Therefore n is the largest integer for which the formula is exact for x^k for all $k = 0, 1, \dots, n$, so the degree of precision is n . $\square$

Problem 5.

$$I = \int_1^2 x \ln x \, dx, \quad f(x) = x \ln x.$$

Derivatives: $f''(x) = 1/x$, $f^{(4)}(x) = 2/x^3$. On $[1, 2]$:

$$\max_{[1,2]} |f''(x)| = f''(1) = 1, \quad \max_{[1,2]} |f^{(4)}(x)| = f^{(4)}(1) = 2.$$

Set tolerance $\varepsilon = 10^{-5}$, $b - a = 1$.

(a) Composite Trapezoidal rule ($h = (b - a)/n$).

$$|E_T| \leq \frac{(b - a) h^2}{12} \max |f''| = \frac{1}{12n^2}.$$

Require $\frac{1}{12n^2} \leq 10^{-5}$, i.e., $n^2 \geq \frac{10^5}{12} \approx 8333.33$, so $n \geq 91.288$. Check $n = 91$: $1/(12 \cdot 91^2) \approx 1.006 \times 10^{-5} > 10^{-5}$ (fails); $n = 92$: $1/(12 \cdot 92^2) \approx 9.85 \times 10^{-6} \leq 10^{-5}$.

$n = 92, \quad h = \frac{1}{92} \approx 0.010870 \quad (E_T \leq 9.85 \times 10^{-6}).$

(b) Composite Simpson's rule (n must be even since subintervals are paired; $h = (b - a)/n$).

$$|E_S| \leq \frac{(b - a) h^4}{180} \max |f^{(4)}| = \frac{2}{180 n^4} = \frac{1}{90n^4}.$$

Require $\frac{1}{90n^4} \leq 10^{-5}$, i.e., $n^4 \geq \frac{10^5}{90} = \frac{10^4}{9} \approx 1111.11$, so $n \geq 5.774$. Check $n = 4$ (even): $1/(90 \cdot 256) \approx 4.34 \times 10^{-5} > 10^{-5}$ (fails); $n = 6$: $1/(90 \cdot 1296) \approx 8.57 \times 10^{-6} \leq 10^{-5}$.

$n = 6, \quad h = \frac{1}{6} \approx 0.166667 \quad (E_S \leq 8.57 \times 10^{-6}).$

(c) Composite Midpoint rule (n even since midpoints come in paired blocks; $h = (b - a)/(n + 2)$, B&F 10th ed. Eq. (4.34)).

$$|E_M| \leq \frac{(b - a) h^2}{6} \max |f''| = \frac{1}{6(n + 2)^2}.$$

Require $\frac{1}{6(n + 2)^2} \leq 10^{-5}$, i.e., $(n + 2)^2 \geq \frac{10^5}{6} \approx 16666.67$, so $n + 2 \geq 129.10$, i.e., $n \geq 127.10$. Check $n = 126$ (even, $n + 2 = 128$): $1/(6 \cdot 128^2) \approx 1.017 \times 10^{-5} > 10^{-5}$ (fails); $n = 128$ ($n + 2 = 130$): $1/(6 \cdot 130^2) \approx 9.86 \times 10^{-6} \leq 10^{-5}$.

$n = 128, \quad h = \frac{1}{130} \approx 0.007692 \quad (E_M \leq 9.86 \times 10^{-6}).$

Problem 6.

With $h_1 = b - a$ (so $n = 1$) and $h_2 = (b - a)/2$ (so $n = 2$), the composite Trapezoidal rule (1) gives:

$$R_{1,1} = \frac{b - a}{2} [f(a) + f(b)],$$

$$R_{2,1} = \frac{b - a}{4} [f(a) + 2f(\frac{a+b}{2}) + f(b)].$$

Richardson's extrapolation:

$$R_{2,2} = R_{2,1} + \frac{1}{3}(R_{2,1} - R_{1,1}).$$

Let $A = f(a)$, $M = f((a + b)/2)$, $B = f(b)$. Then:

$$R_{2,1} - R_{1,1} = \frac{b - a}{4}(A + 2M + B) - \frac{b - a}{2}(A + B) = \frac{b - a}{4}(-A + 2M - B),$$

$$\frac{1}{3}(R_{2,1} - R_{1,1}) = \frac{b - a}{12}(-A + 2M - B),$$

$$R_{2,2} = \frac{b - a}{4}(A + 2M + B) + \frac{b - a}{12}(-A + 2M - B)$$

$$= \frac{b - a}{12}[3(A + 2M + B) + (-A + 2M - B)]$$

$$= \frac{b - a}{12}(2A + 8M + 2B) = \frac{b - a}{6}(A + 4M + B).$$

$R_{2,2} = \frac{b - a}{6}[f(a) + 4f(\frac{a+b}{2}) + f(b)].$

With $h = (b - a)/2$, this is exactly Simpson's rule:

$$R_{2,2} = \frac{h}{3}[f(a) + 4f(\frac{a+b}{2}) + f(b)].$$

Problem 7.

The IVP is $y' = [\exp(t) y]^2 = e^{2t} y^2$, $y(0) = -1$, $0 \leq t \leq 1$, with

$$f(t, y) = e^{2t} y^2, \quad D = \{(t, y) : 0 \leq t \leq 1, -\infty < y < \infty\}.$$

Continuity. f is a polynomial in y multiplied by a continuous function of t , hence continuous on D . ✓

Theorem 5.3 cannot establish Lipschitz. Compute $\partial f / \partial y = 2e^{2t}y$. On D the variable y ranges over all of $\mathbb{R}$, so

$$\sup_{(t,y) \in D} \left| \frac{\partial f}{\partial y} \right| = \sup_{0 \leq t \leq 1} \sup_{y \in \mathbb{R}} 2e^{2t}|y| = +\infty.$$

Theorem 5.3 (which gives a Lipschitz constant $L = \sup_D |\partial f / \partial y|$) requires this supremum to be finite, so it does not yield a Lipschitz constant for f on D . But Theorem 5.3 is only sufficient; we must check the Lipschitz condition directly.

Direct verification that f is not Lipschitz on D . Suppose, for contradiction, that there exists $L > 0$ with $|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$ for all $(t, y_1), (t, y_2) \in D$. Take $t = 0$, $y_2 = 0$, and any $y_1 = M > 0$:

$$\frac{|f(0, M) - f(0, 0)|}{|M - 0|} = \frac{|M^2 - 0|}{M} = M.$$

This must be $\leq L$, but M is unrestricted in D . Contradiction. Hence f does **not** satisfy a Lipschitz condition on D , so the hypothesis of Theorem 5.4 fails.

Theorem 5.4 **cannot** be applied to (2) on $D = \{0 \leq t \leq 1, y \in \mathbb{R}\}$.

Remark. A unique solution $y(t) = -2/(1 + e^{2t})$ does exist (Theorem 5.4 is only sufficient). Verification:

$$y'(t) = -2 \cdot \frac{-2e^{2t}}{(1 + e^{2t})^2} = \frac{4e^{2t}}{(1 + e^{2t})^2} = e^{2t} \left(\frac{-2}{1 + e^{2t}} \right)^2 = e^{2t} y^2 = [\exp(t) y]^2,$$

and $y(0) = -2/2 = -1$. Since $y(0) = -1$ and y increases monotonically toward 0 on $[0, 1]$, $|y(t)| \leq 1$ throughout. Restricting D to a strip $\{|y| \leq M\}$ would render $\partial f / \partial y$ bounded, and Theorem 5.4 would then apply.

Problem 8.

$y' = t(y + 1)$, $y(0) = 0$, $0 \leq t \leq 1$, $h = 1/3$.

Euler's method: $w_{i+1} = w_i + h \cdot f(t_i, w_i) = w_i + h t_i(w_i + 1)$, with $t_i = ih$.

$w_0 = 0,$	$t_0 = 0,$
$w_1 = w_0 + \frac{1}{3} \cdot 0 \cdot (0 + 1) = 0,$	$t_1 = \frac{1}{3},$
$w_2 = w_1 + \frac{1}{3} \cdot \frac{1}{3} \cdot (0 + 1) = \frac{1}{9},$	$t_2 = \frac{2}{3},$
$w_3 = w_2 + \frac{1}{3} \cdot \frac{2}{3} \cdot (\frac{1}{9} + 1) = \frac{1}{9} + \frac{2}{9} \cdot \frac{10}{9} = \frac{1}{9} + \frac{20}{81} = \frac{9+20}{81} = \frac{29}{81},$	$t_3 = 1.$

i	t_i	w_i	$y(t_i) = e^{t_i^2/2} - 1$
0	0	0	0
1	1/3	0	$e^{1/18} - 1 \approx 0.05709$
2	2/3	1/9	$e^{2/9} - 1 \approx 0.24860$
3	1	29/81	$e^{1/2} - 1 \approx 0.64872$

$$w_0 = 0, \quad w_1 = 0, \quad w_2 = \frac{1}{9}, \quad w_3 = \frac{29}{81}.$$